CAPSTONE PROJECT PROPOSAL

IRONCLAD GROUP

Healthcare Supply Chain Intelligence Platform for Optimizing Medical Commodity Distribution in Kenya

1. Team Members

Name	Role
DEBORAH OMAE	Data Engineering Lead — ETL pipelines, data structuring, analytics-ready healthcare supply chain datasets DevOps Lead - deployment readiness documentation
CAMILA AOKO	Modelling & ML Co-Lead — exploratory analysis, demand forecasting, stockout-risk modelling QA & Documentation Lead — system testing, UAT
MICHAEL MUNGA	Modelling & ML Co-Lead — exploratory analysis, demand forecasting, allocation recommendation modelling
BRIAN SIGEI	Dashboard & Visualization Lead — KEMSA, county, and facility-level stakeholder dashboards
EDNAH KIMANI	

2. Problem Statement

Kenya's public healthcare system relies on the Kenya Medical Supplies Authority (KEMSA) to procure, warehouse, and distribute essential medical commodities to public health facilities across all 47 counties, yet many facilities still experience stockouts, overstocking, and expiry of unused commodities due to limited visibility into real demand and inventory requirements. Most facilities continue to depend on manual, paper-based bin cards and periodic reporting, which delays understanding of real-time consumption patterns and prevents accurate demand forecasting, leaving some facilities overstocked with commodities that expire unused while others run short of essentials such as antibiotics, IV fluids, and maternal health supplies, pushing patients toward private pharmacies, raising out-of-pocket costs, and undermining Kenya's universal healthcare objectives. We propose a healthcare supply chain intelligence platform that integrates facility, warehouse, and historical consumption data to forecast demand, flag facilities at risk of stockouts before shortages occur, detect overstock situations likely to expire, and recommend optimized allocation and redistribution shifting KEMSA's supply chain from a reactive ordering model to a proactive, data-driven one that ensures the right commodity reaches the right facility in the right quantity at the right time.

3. Domain & Context

Industry: Public Health & Pharmaceutical Supply Chain, Kenya.

KEMSA procures, warehouses, and distributes essential medical commodities to public health facilities across all 47 counties, coordinating with county health departments and individual facility managers who currently track inventory largely through manual bin cards and periodic reporting rather than real-time systems. Although Kenya Pipeline Company (KPC) does not handle medical commodities, its experience running large-scale logistics monitoring — telemetry, inventory tracking, and demand planning across a national petroleum distribution network, offers a transferable model for healthcare logistics. This creates a strategic opening for KPC to act as a Digital Logistics Intelligence Partner: supplying supply chain analytics platforms to government institutions, contributing industrial IoT and monitoring expertise, supporting data infrastructure through its fiber connectivity network, and helping develop public-sector logistics intelligence solutions — positioning KPC as a digital transformation partner in national critical supply chains that extends well beyond petroleum transportation.

4. Data Source

Real-time, facility-level KEMSA inventory and consumption data is not openly published, so the project draws on a layered, realistic data strategy:

- Primary: facility and warehouse-level records. Inventory, dispensing/consumption, and distribution data structured to reflect how KEMSA and county health facilities track medical commodities, including antibiotics, IV fluids, and maternal health supplies.
- Reference: public health reporting structures. Kenya's Health Information System (KHIS/DHIS2) county-level reporting patterns used to ground realistic demand and consumption assumptions across the 47 counties.

- Validation: known supply chain patterns. Reported stockout, wastage, and expiry trends from KEMSA and county health departments used to sanity-check model outputs before pilot deployment.

5. Success Metrics

Metric	Target
Demand Forecast Accuracy	**≥ 80%**
Stockout Reduction (Pilot Counties)	**Demonstrated reduction vs. baseline**
Expiry / Wastage Reduction	**Demonstrated reduction vs. baseline**
Early-Warning Lead Time for At-Risk Facilities	**Sufficient window for re-supply before stockout**
Dashboard Deployment	**Functional views for KEMSA, county, and facility levels**
Pilot Coverage	**Selected counties representing distinct operational environments**
National Scale-Up Readiness	**Documented expansion plan across all 47 counties**

Additional evaluation criteria: dashboard usability across KEMSA, county, and facility stakeholders; ease of adoption by facility staff; documentation quality; and pilot-to-national deployment readiness.

6. Implementation Roadmap

Phase	Lead	Focus	Deliverable
1	Project Team	Stakeholder Engagement & Requirements Gathering	Defined KPIs, data sources, and supply chain challenges from KEMSA, county health departments, and facilities
2	Deborah Omae	Data Engineering & Platform Development	ETL pipelines and analytics-ready healthcare supply chain dataset
3	Camila Aoko & Munga	Machine Learning Development	Demand forecasting models, stockout-risk prediction, allocation recommendations

4	Brian Sigei	Dashboard & Visualization Development	Dashboards for KEMSA executives, county health departments, and facilities
5	Ednah Kimani	Testing, Documentation & Deployment Readiness	UAT results and deployment documentation
6	Full Team	Pilot Deployment & Scaling	Pilot results in selected counties and a national scaling plan across all 47 counties

Conclusion

This solution transforms healthcare supply chain management from a reactive ordering system into a predictive intelligence ecosystem. By combining data engineering, artificial intelligence, and industrial logistics principles, the platform creates value for KEMSA, health facilities, patients, solution providers, and strategic partners such as KPC — first piloted in selected counties representing different operational environments, then scaled nationally across all 47 counties.